

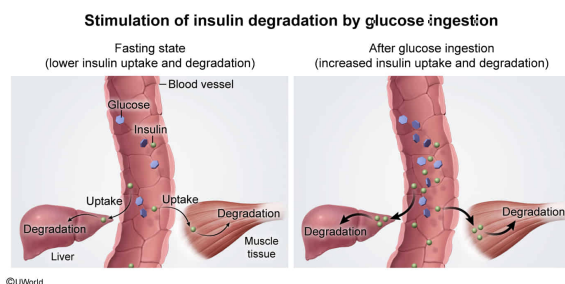
Insulin molecules are degraded after receptor binding and internalization into target cells. Compared to the fasting state (low glucose uptake by target cells), the insulin degradation rate in a nondiabetic individual shortly after consuming a glucose solution will most likely be:

- ☐ A. approximately the same. [3%]
- ✓ ☒ B. increased. [62%]
- ☐ C. decreased. [22%]
- ☐ D. decreased, but higher than in a diabetic individual. [10%]

Correct

61% answered correctly

### Explanation:



Glucose is an essential molecule within the body, serving both as an **energy substrate** used for ATP generation and as a source of necessary metabolic intermediates (eg, oxaloacetate for the oxidation of **macronutrient metabolites** in the **Krebs cycle**). In the **fasting state**, the circulating blood glucose level is maintained by **glycogen stores** in the **liver** (ie, glycogenolysis) as well as from **gluconeogenesis**. Conversely, after consumption of a carbohydrate-containing meal, the blood glucose level increases due to glucose absorption into the bloodstream.

Following carbohydrate consumption, glucose uptake from the circulation is facilitated (particularly in nondiabetic individuals) by insulin secretion from the **pancreas**. The secreted insulin molecules bind to the insulin receptor in target tissues, causing translocation of glucose transporters to the plasma membrane. This activity results in markedly higher glucose uptake rates than those observed under fasting conditions.

The question states that the glucose uptake rate is low under fasting conditions, and insulin molecules are degraded shortly after internalization into target cells. The question also asks for the insulin degradation rate shortly after a nondiabetic individual consumes a glucose solution as compared to the rate in the same individual under fasting conditions.

In a nondiabetic individual, **drinking a glucose solution increases insulin binding** to receptors on target cell plasma membranes, followed by **internalization and destruction** of the insulin. Therefore, the **insulin degradation rate** in a nondiabetic individual **shortly after consuming a glucose solution will be increased** (*not* approximately the same or

decreased) relative to the fasting state (**Choices A, C, and D**).

**Educational objective:**

The pancreas secretes insulin in response to increased blood glucose concentrations, and the secreted insulin binds to receptors on target cells to promote glucose transport.

**Time spent:**0 sec

**Subject:**  
Biology

**QID:**404103

**System:**  
Endocrine and Nervous Systems